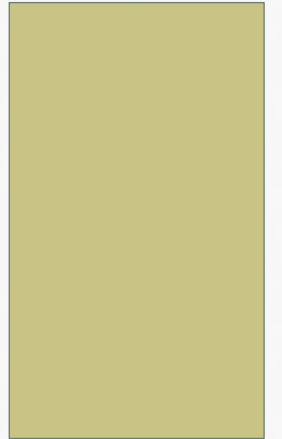


Class II Cavity Preparation



LEARNING OUTCOMES

By the end of this lecture, students should be able to

- Describe steps and outlines the objectives of class II cavity preparation
- Contrast different rotary and hand instruments used in class II cavity preparation

•Class II Preparation:

Preparation on the proximal surfaces of posterior teeth are class II.



CLASS II CAVITY PREPARATION



Stages of Cavity Preparation



❑ INITIAL STEPS OF CAVITY PREPARATION

OUTLINE FORM AND INITIAL DEPTH



PRIMARY RESISTANCE FORM



PRIMARY RETENTION FORM



CONVENIENCE FORM

❑ FINAL STEPS OF PREPARATION

REMOVAL OF ANY REMAINING INFECTED DENTIN OR OLD RESTORATION



PULP PROTECTION



SECONDARY RESISTANCE AND RETENTION FORM



FINISHING THE EXTERNAL WALLS



CLEANING, INSPECTING ,SEALING

CLASS II CAVITY PREPARATION

- The tooth preparation for a Class II composite restoration may be either a
 1. Conventional
 2. Modified preparation design.
- The modified design is for smaller restorations, generally using smaller burs and resulting in a preparation form that is more rounded, less boxlike, and less uniform in extension or depth

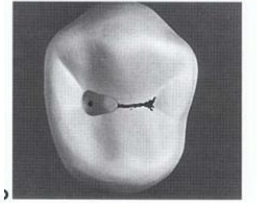


CONVENTIONAL CLASS II TOOTH PREPARATION

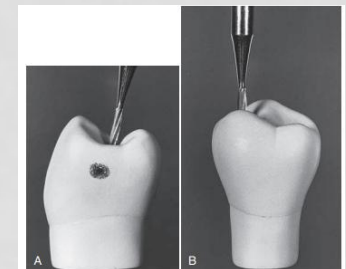
The tooth preparation has two components—

- 1. The occlusal step portion
- 2. Proximal box portion.

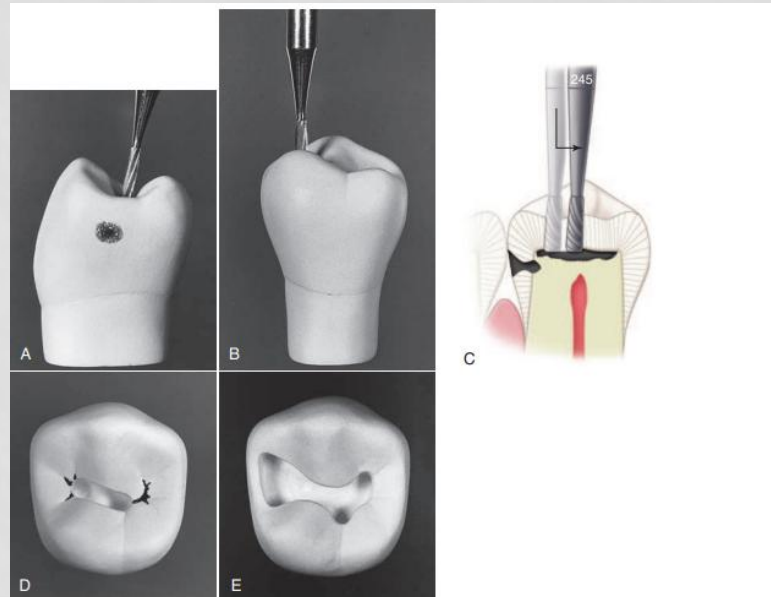
Occlusal Step



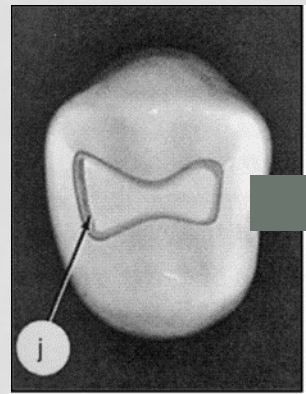
- The occlusal portion of the Class II preparation is prepared similarly for the Class I preparation.
- Using high speed with air-water spray, the operator enters the pit nearest the involved proximal surface with a punch cut using a No. 245 bur or a 330 bur (No. 330 bur is a smaller and pear-shaped version of the No. 245 bur)
- The bur should be rotating, when it is applied to the tooth and should not stop rotating until removed.



- The bur is positioned parallel with the long axis of the tooth.
- While maintaining the same depth and bur orientation, the bur is moved to extend the outline to include the carious central fissure

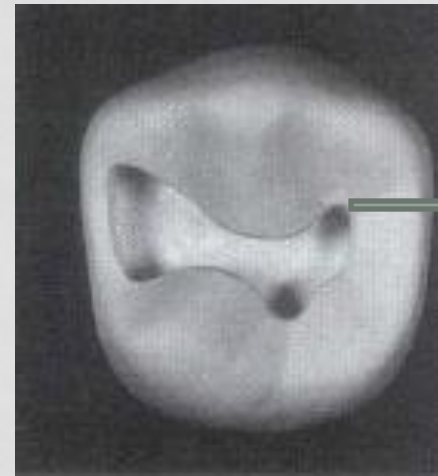


- If only one proximal surface is being restored, the opposite marginal ridge dentinal support should be maintained.
- The pulpal floor is prepared with the instrument to a depth of 1.5mm, as measured from the central groove.
- The pulpal floor is relatively flat

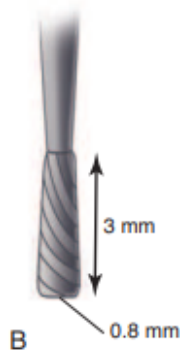


Marginal ridge

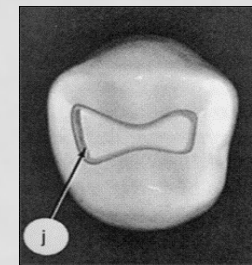
- The distal pit (Opposite) area provides dovetail retention form, which may prevent mesial displacement of the completed restoration.
- A dovetail outline form in the distal pit is not required if radiating fissures are not present.



Dovetail



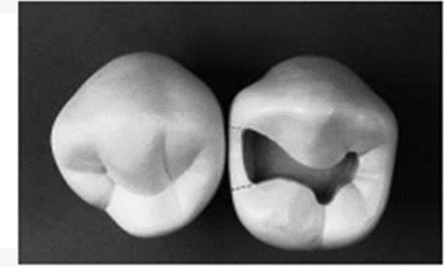
- The isthmus width should be as narrow as possible, preferably no wider than one quarter the intercusp distance. ($\frac{1}{4}^{\text{th}}$ of the intercusp distance)
- Ideally the preparation should be the width of the No. 245 bur (0.8 mm). Narrow restorations provide a greater length of clinical service.
- Generally the amount of remaining tooth structure is more important to restoration longevity than the restorative material used.



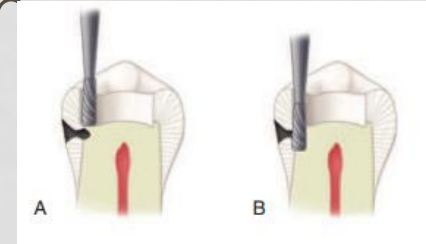
- While maintaining the established pulpal depth and with the bur parallel to the long axis of the tooth, the preparation is extended mesially (In case of Mesioocclusal preparation) stopping approximately 0.8 mm short of the marginal ridge
- The occlusal portion in this region is made slightly wider faciolingually than in the Class I preparation, because additional width is necessary for the proximal box.



PROXIMAL BOX

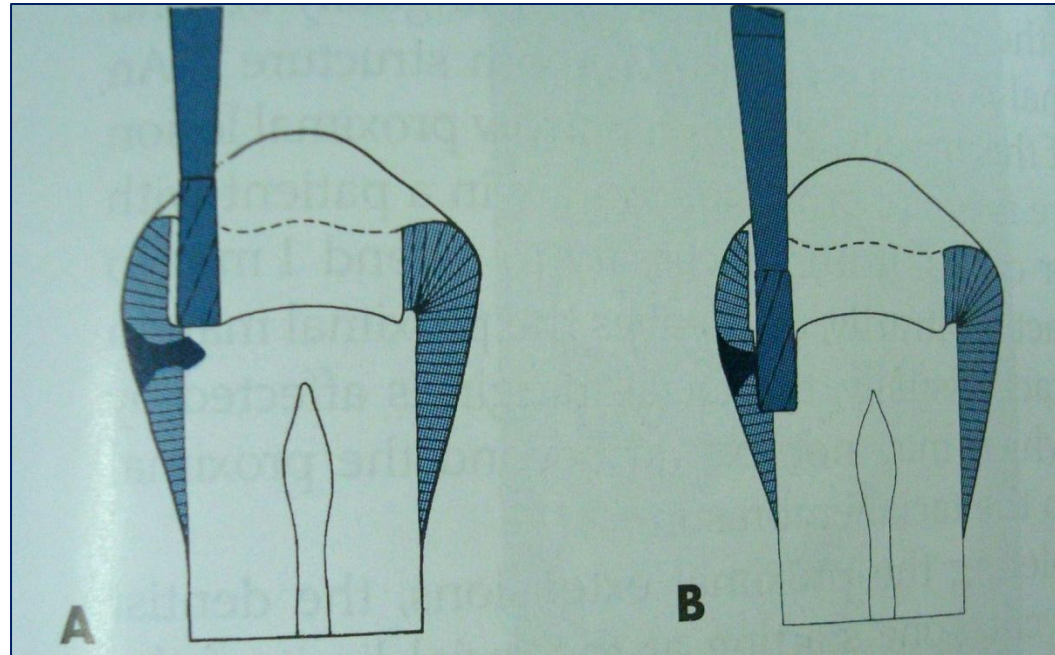


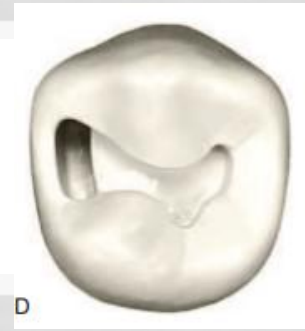
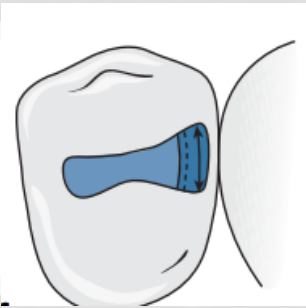
- Before initiating the proximal outline form (i.e., the proximal “box”), the operator should visualize the desired final location of the facial and lingual walls relative to the contact.
- While maintaining the same orientation of the bur in the pulpal floor next to the remaining mesial marginal ridge, the bur is allowed to cut a ditch gingivally two thirds at the expense of enamel and one third at the expense of dentin. The 0.8 mm of bur end cuts approximately 0.5 to 0.6 mm into enamel and 0.2 to 0.3 mm into dentin.



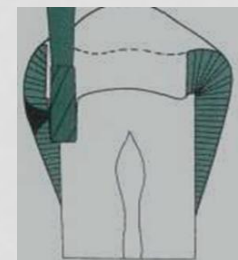
PROXIMAL DITCH

- A. Bur position to start the proximal ditch.
- B Proximal ditch extended gingivally to the desired level of gingival seat

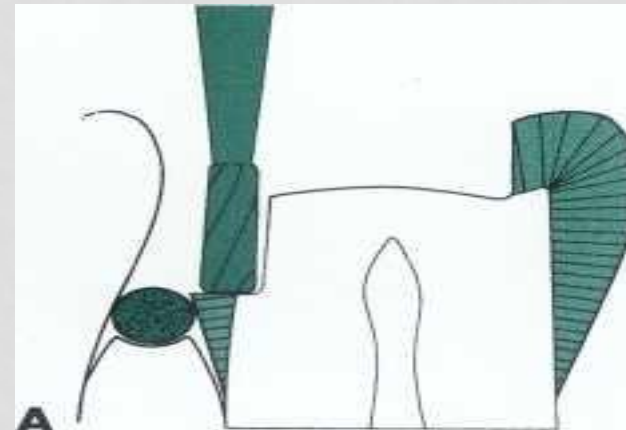




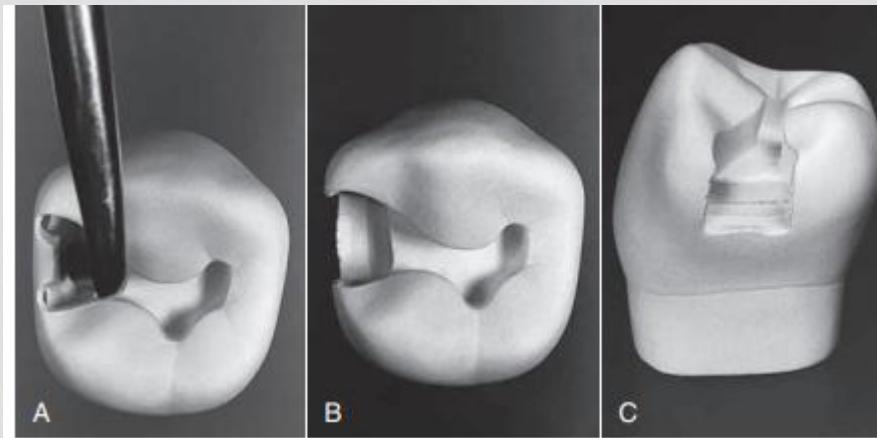
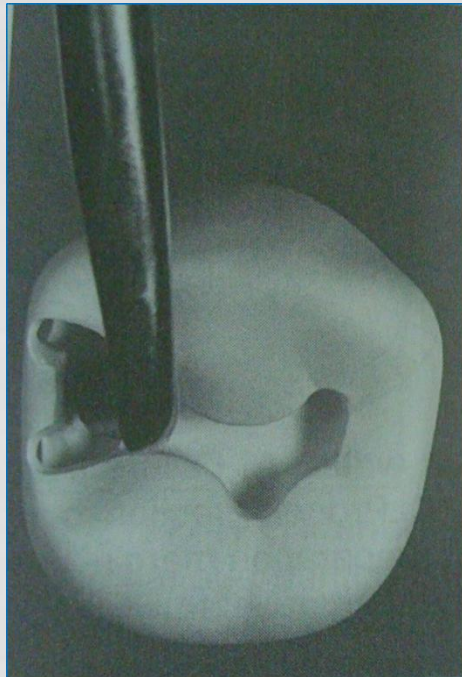
- Pressure is directed gingivally and lightly toward the mesial surface to keep the bur against the proximal enamel, while the bur is moved facially and lingually. The faciolingual cutting motion follows the DEJ (Dentino enamel Junction) and is usually in a slightly convex arc .
- The axial wall follows the faciolingual contour of the proximal surface and the DEJ and is convex
- The proximal depth is extended gingivally just beyond the caries lesion or the proximal contact, whichever is greater



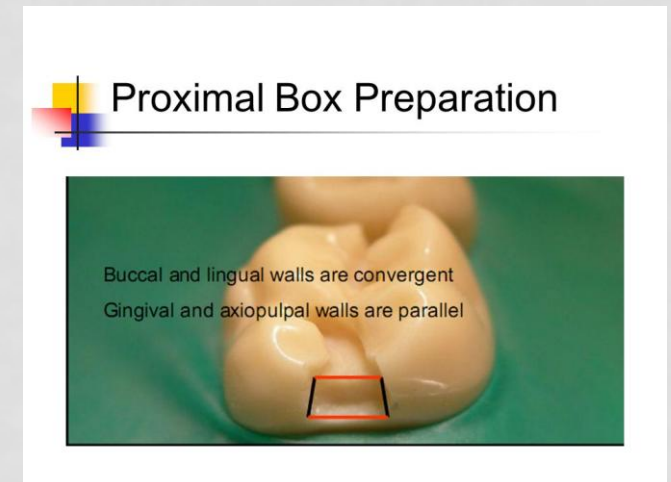
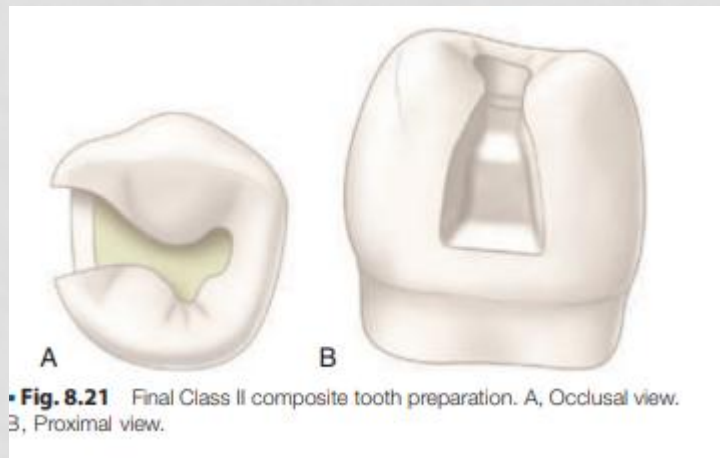
- To protect the gingiva and the rubber dam when extending the gingival seat gingivally, a wooden wedge may be placed in the gingival embrasure to depress the soft tissue and rubber dam.
- Wedging increases the interproximal space and limits the potential for iatrogenic damage of adjacent tooth (or restoration) surfaces.

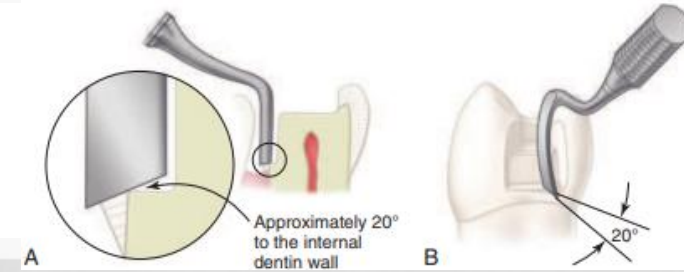
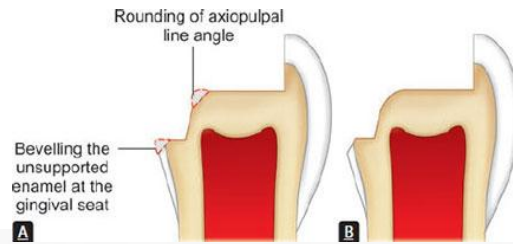
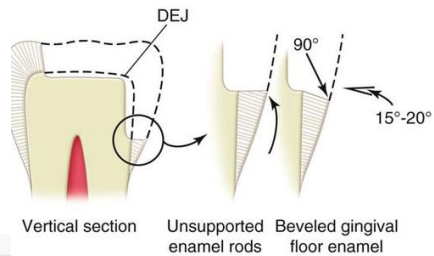


- The weakened enamel along the gingival wall is removed by using the enamel hatchet in a scraping motion

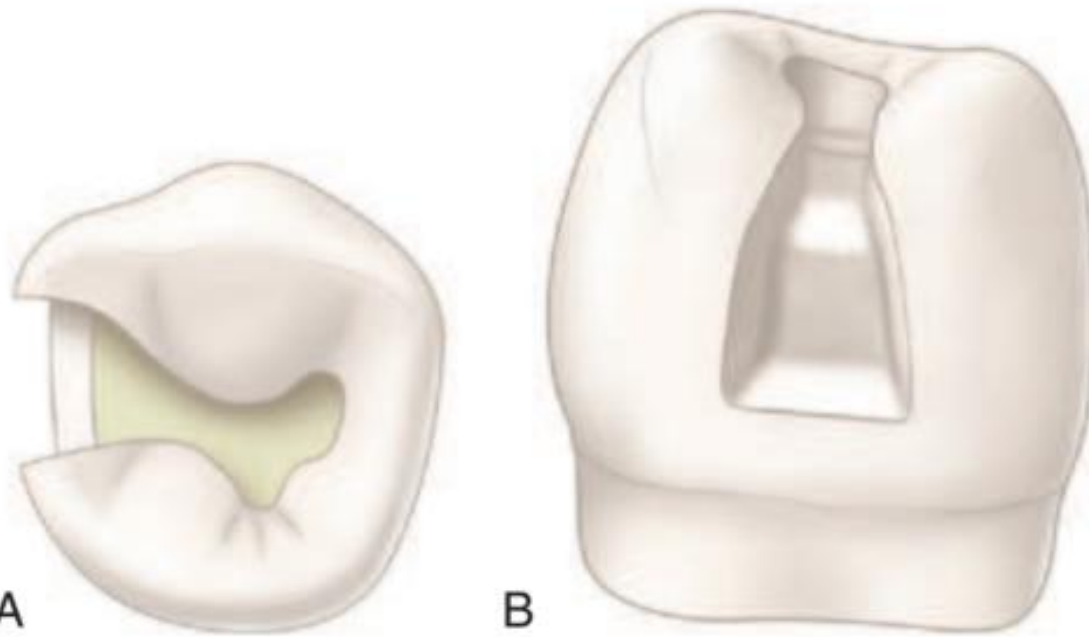


- The proximal ditch cut may diverge gingivally (i.e., converge occlusally) to ensure that the faciolingual dimension at the gingival aspect is greater than at the occlusal aspect
- The shape of the No. 245 bur will provide this . The gingival divergence contributes to the retention form





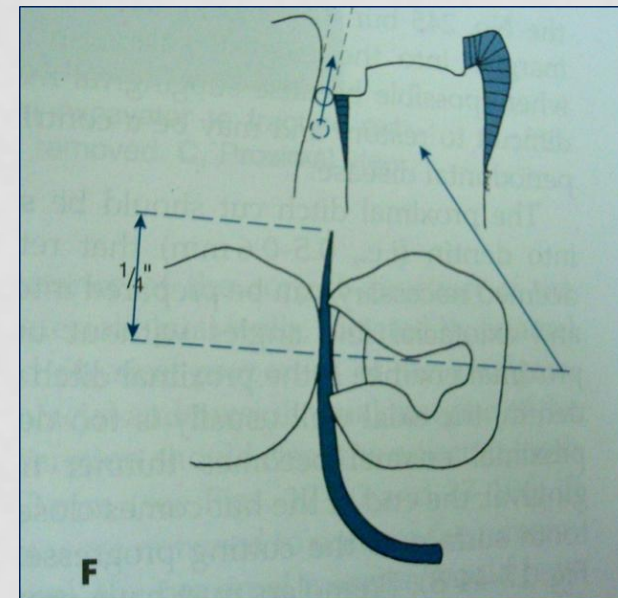
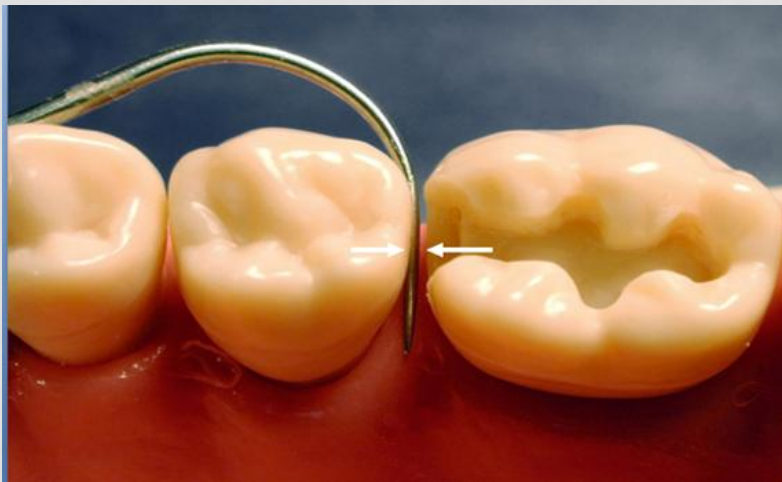
- A sharp gingival margin trimmer is used to make Cavo surface bevel at gingival cavosurface margin. This results in enamel gingival margin has a 20-degree gingival declination. (In a Class II amalgam Preparation)
- In a class II composite preparation ,because the composite is retained primarily by micromechanical adhesion, no secondary preparation retention features are necessary.
- Bevels are not recommended in class II composite preparation along the gingival cavosurface margins of the proximal box; however, it is necessary remove brittle unsupported enamel rods along the margins because of the gingival orientation of the enamel rods.



• **Fig. 8.21** Final Class II composite tooth preparation. A, Occlusal view. B, Proximal view.



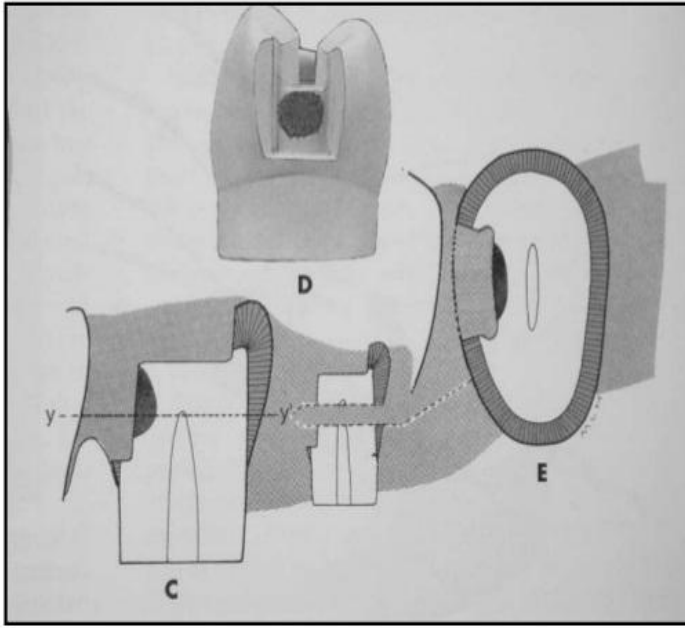
- Gingival and pulpal floors should be flat to obtain resistance form
- Ideally the gingival margin should clear the adjacent tooth by only 0.5mm ,which may be measured with the fine of explorer.



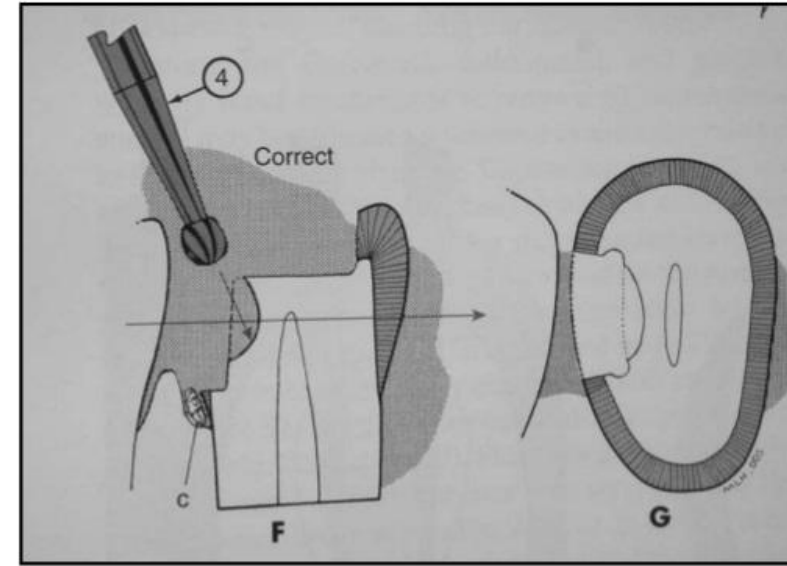
FINAL TOOTH PREPARATION

Removal of Infected Carious Dentin and Pulp Protection.

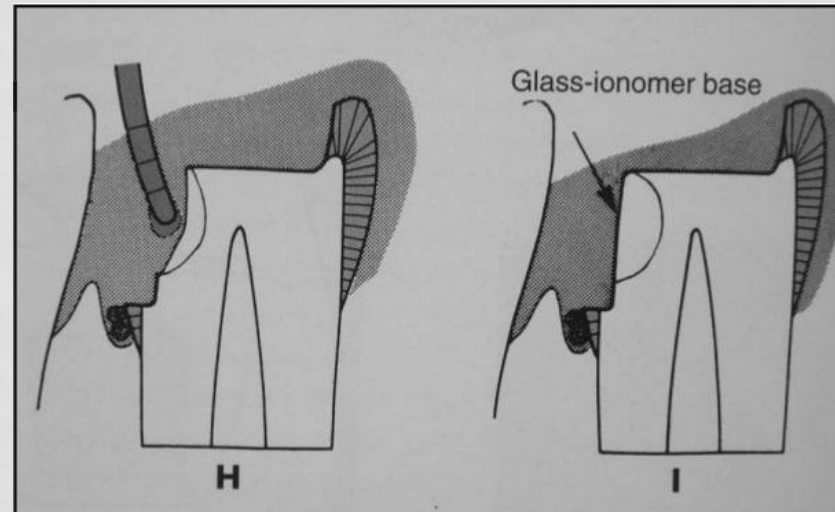
- After the initial preparation has been completed, the dentist evaluates the internal walls of the preparation visually and tactilely (with an explorer) for indications of remaining carious dentin.



Remaining caries on the axial wall



Removing the remaining infected dentin with
no. 2 or 4 round bur



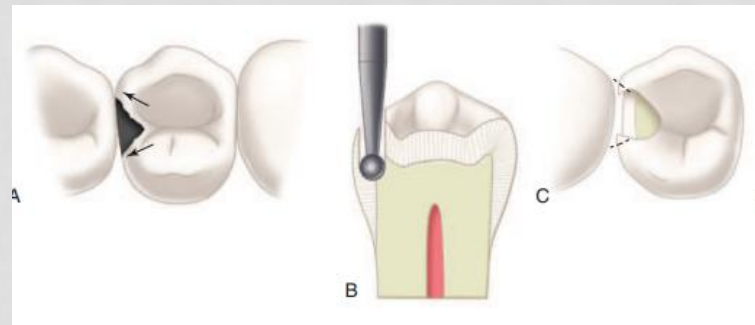
Insertion of suitable base and completed base.

- A slowly revolving round bur (No. 2 or No.4) or spoon excavator is used to remove the carious infected dentin
- Light-cured glass-ionomer cement may be mixed and applied with a suitable applicator to these moderately deep excavated regions to form of the ideally cavity surface

MODIFIED CLASS II COMPOSITE PREPARATION

First design – Conservative design

- For small, initial restorations, an even more conservative preparation design may be used.
- A small round or inverted cone bur may be used for this preparation to scoop out the carious or faulty material.
- To prevent damage to the adjacent teeth and promote initial interproximal separation, wedges with or without stainless steel barriers may be utilized.

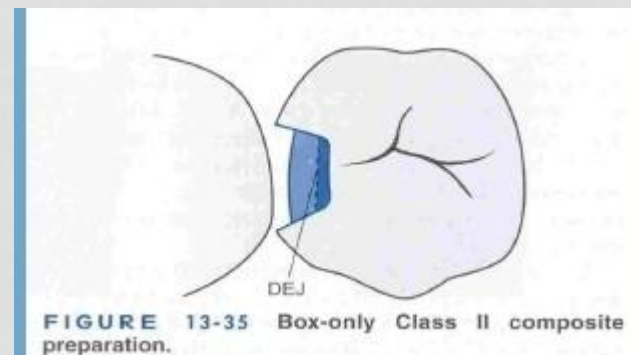


- Objective of conservative design of a class II preparation is to remove the caries lesion or defect conservatively, as well as any friable tooth structure.
- This scooped appearance occurs on the occlusal and the proximal portions.
- The pulpal and axial depths are dictated only by the depth of the lesion and are not uniform.



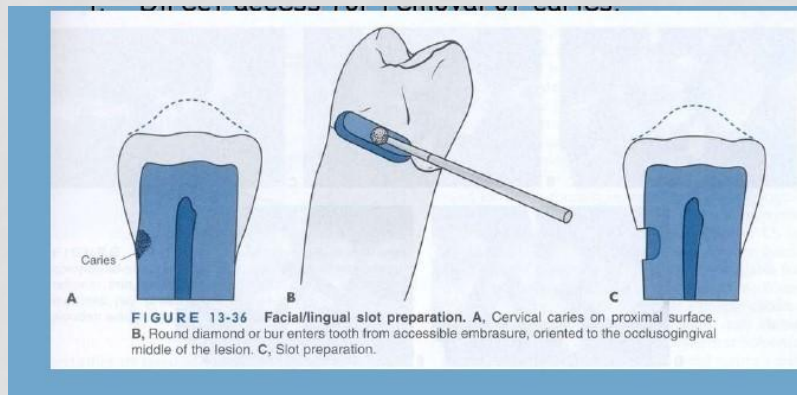
- **Second design – Box Design**

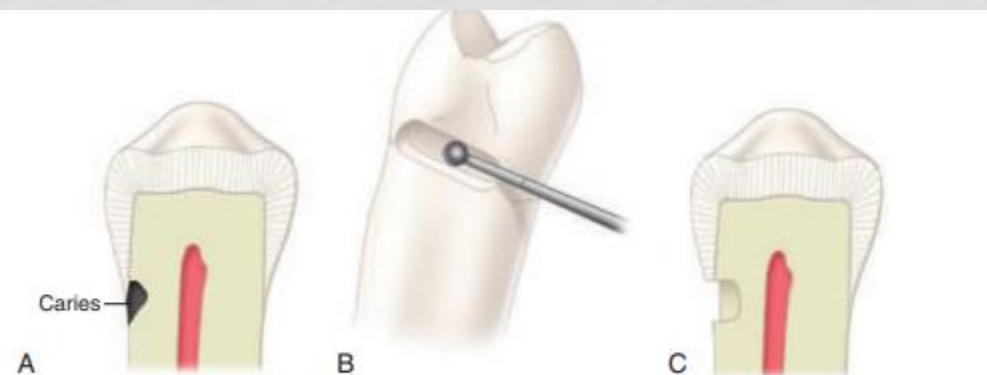
- This design is indicated when only the proximal surface is faulty, with no lesions on the occlusal surface.
- A proximal box is prepared with an inverted cone or round bur, held parallel to the long axis of the tooth crown.



- The bur is extended through the marginal ridge in a gingival direction. The axial depth is prepared 0.2mm inside the DEJ
- The facial, lingual, and gingival extensions are dictated by the fault or caries. No beveling or secondary retention is indicated.

- **Third design – Slot design**
- Here a lesion or caries is accessed directly from the proximal surface, rather than through the marginal ridge .
- Usually, a small round diamond or bur is used to gain access to the lesion.
- Most commonly used if adjacent teeth is missing





• **Fig. 8.15** Facial or lingual slot preparation. A, Cervical caries on the proximal surface. B, The round diamond or bur enters the tooth from the accessible embrasure, oriented to the occlusogingival middle of the lesion. C, Slot preparation.

REFERENCES

- Art and science operative dentistry 5 th edition
Page no – 574 to 589
- Art and science operative dentistry 7 th edition
Page no – 229 to 236